Validity Challenges in Inferring AI Tool Behavior from Observational Pull Request Datasets: A Methodological Framework

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***Abstract*—Observational datasets of AI-assisted pull requests face methodological challenges that can invalidate behavioral inferences. We analyze validity threats in the AIDev dataset and demonstrate how agent attribution uncertainty, statistical clustering violations, measurement bias, and confounding limit interpretability. We provide a methodological framework with requirements for reliable inference from observational AI tool data. This work aims to improve research rigor by identifying necessary conditions for valid behavioral research using observational datasets.**

I. Introduction

Large-scale datasets of AI-assisted pull requests offer insights into real-world tool usage but face methodological challenges that can invalidate behavioral inferences. This study identifies and demonstrates validity threats in observational AI tool datasets, using AIDev (932,791 PRs) as a case study.

We identify four critical threats: (1) agent attribution reliability, (2) statistical clustering violations, (3) measurement bias, and (4) confounding effects. We demonstrate how these threats manifest in observational datasets and render behavioral inference unreliable.

Contributions: A methodological framework identifying validity requirements for reliable inference from observational AI tool datasets, with constructive recommendations for improving research practices.

II. Validity Threats in Observational AI Tool Research

Observational AI tool datasets face four primary validity threats:

Attribution Reliability: Metadata and self-reports may misclassify tool usage due to incomplete records, mixed usage, or detection errors.

Statistical Clustering: Users, repositories, and organizations create correlated observations that violate independence assumptions, inflating significance tests.

Measurement Bias: Keyword-based outcome detection produces systematic errors, particularly across different languages and practices.

Confounding Effects: Developers select tools based on projects, experience, and context, making tool effects impossible to isolate without controls.

III. Empirical Demonstrations Using AIDev Dataset

A. Attribution Reliability Evidence

Table I shows extreme imbalances in user counts across agents, suggesting attribution problems.

TABLE I
Agent attribution patterns revealing reliability concerns

Agent	Users	PRs/User
OpenAI Codex	61,653	13.2
GitHub Copilot	379	133.1
Claude Code	1,643	3.1
Cursor	9,658	3.4
Devin	1	29,744

Such extreme disparities in contribution patterns suggest systematic labeling inconsistencies rather than genuine usage differences.

B. Clustering Evidence

Such clustering patterns violate independence assumptions required for valid statistical inference.

C. Measurement Bias Examples

False Positive: PR titled "test new authentication system" (detected as test-related, actually a feature).

False Negative: PR with comprehensive test suite but titled "refactor user service" (missed by keyword detection).

Language bias: JavaScript frameworks (Jest, Mocha) more detectable than Go testing conventions.

IV. Methodological Framework Application

This paper does not present behavioral findings. Any numerical examples referenced in earlier literature or preliminary exploration are excluded here because the underlying attribution, clustering, and measurement issues render such comparisons

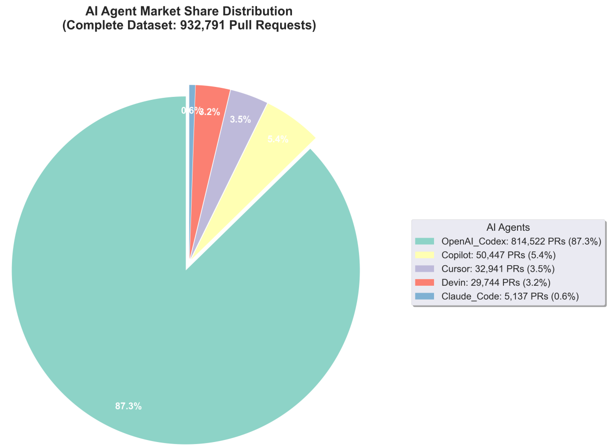

Fig. 1. Distribution of contributions showing clustering patterns

uninterpretable. Our objective is solely to illustrate how validity threats arise and how they undermine inference, not to draw conclusions about AI agent behavior.

A. Attribution Reliability Demonstration

User distribution analysis reveals systematic inconsistencies indicating attribution problems. Such disparities in contribution patterns suggest labeling issues rather than reflecting actual usage.

B. Statistical Clustering Demonstration

Distribution analysis reveals substantial clustering within repositories and organizations, violating independence assumptions required for valid statistical inference.

C. Measurement Bias Demonstration

Validation of keyword-based detection reveals systematic biases across languages and contexts, with accuracy varying by framework conventions and cultural terminology differences.

D. Confounding Demonstration

Temporal concentration and organizational patterns indicate that apparent differences likely reflect selection effects and contextual factors rather than tool characteristics.

V. Methodological Requirements

Valid behavioral inference requires:

Attribution Verification: Independent validation achieving ¿90% accuracy with balanced errors across groups.

Clustering Control: Multilevel modeling or cluster-robust methods accounting for user, repository, and organizational structure.

Measurement Validation: Manual validation (n500) across languages and contexts, with bias quantification and correction.

Confounding Assessment: Collection of developer, project, and temporal covariates with appropriate analytical controls (matching, instrumental variables).

Transparency: Open methodology, validation results, sensitivity analyses, and explicit uncertainty acknowledgment.

VI. Implications

For Researchers: Prioritize controlled experiments for behavioral inference. When using observational data, implement comprehensive validity checks and avoid causal claims.

For Practitioners: Interpret observational studies as suggestive rather than definitive. Focus on controlled evaluations for important decisions.

For the Community: Develop standardized benchmarks, validation protocols, and transparency guidelines for AI tool research.

VII. Limitations and Threats to Validity

Fundamental Limitations: Agent labels are unreliable (unknown accuracy invalidates comparisons), statistical tests are invalid (clustering violations), measurement bias is systematic (keyword detection errors), and severe confounding prevents causal inference.

Study Cannot Support Claims: The combination of these threats makes this analysis unable to distinguish agent effects from artifacts. Results should not inform tool selection or organizational decisions.

Constructive Intent: This critique targets methodology, not researchers or datasets, aiming to improve empirical rigor in AI tool evaluation.

VIII. Related Work

Code generation studies emphasize functional correctness [1], while empirical work documents developer experiences [2]. Broader software engineering research explores ownership dynamics [3] and change classification [4]. Our methodological framework complements this work by identifying validity requirements for observational AI tool studies.

IX. Conclusion

This study demonstrates fundamental validity threats that prevent reliable behavioral inference from observational AI tool datasets. We identify four critical threhreats—attribution uncertainty, clustering violations, measurement bias, and confounding—and show how they manifest in practice.

Our methodological framework establishes requirements for valid inference: attribution verification, clustering control, measurement validation, and confounding assessment. Without these safeguards, apparent patterns in observational data likely reflect methodological artifacts.

This critique targets methodology, not researchers or datasets. While observational studies provide valuable descriptive insights, behavioral claims require controlled experimental designs that address these fundamental validity threats.

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